



YISHUN INNOVA JUNIOR COLLEGE
JC2 PRELIMINARY EXAM
Higher 2

NAME

ANSWERS

INDEX NO

CG

BIOLOGY

9744/02

Paper 2 Structured Questions

29 June 2021

Candidates answer on the Question Paper.

2 hours

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name, index no. and CG on this cover page.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer all questions in the spaces provided on the Question paper.

Section B

Answer any one question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

At the end of the examination, fasten all your work securely together.

For Examiner's Use	
Section A	
1	14
2	10
3	12
4	11
5	12
6	12
7	14
8	10
9	5
Total	100

This document consists of **25** printed pages and **3** blank pages.

Answer **all** questions.

1 (a) Describe the features of embryonic stem cells.

1. **unspecialized / undifferentiated i.e. no specialized cellular structures thus no specific functions;**

2. **are pluripotent**

will differentiate into 3 germ layers (endoderm, mesoderm, ectoderm) to give rise to all organs in an adult organism;

3. **with unlimited/long term self-renewal capacity**

to undergo indefinite division via mitosis;

[3]

(b) Fig. 1.1 shows the changes in energy during the progress of an uncatalysed reaction.

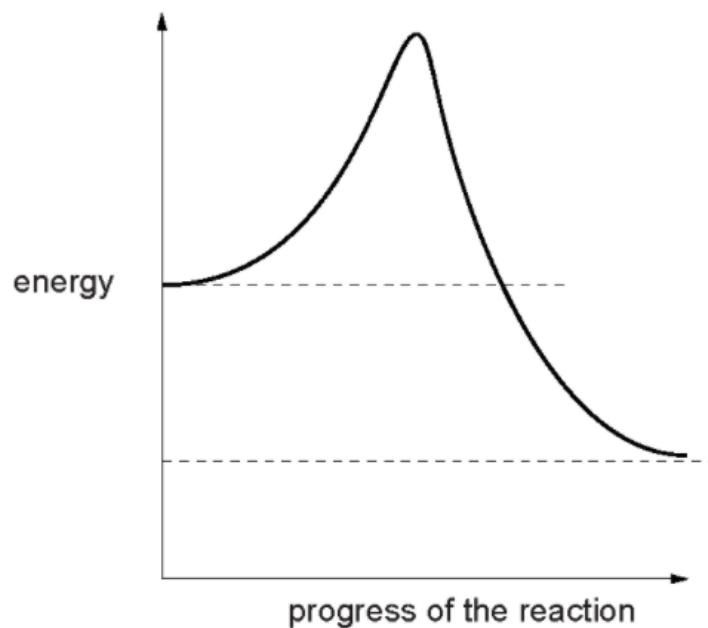


Fig. 1.1

(i) Draw on Fig. 1.1 a curve to show changes in energy during the progress of the same reaction when catalysed by an enzyme.

[2]

Marking points:

1. **one lower peak inside line than uncatalysed;**

2. **starts and finishes at, dotted lines / same energy levels as uncatalysed;**

- (ii) State the term given to the energy level that must be overcome before a reaction can progress.

Activation energy;

[1]

- (c) Sox2, Oct4 and Nanog are transcription factors involved in maintaining the potency of stem cells. When E3 ligase acts on them, differentiation of stem cells occur.

Some of the key events of differentiation of stem cells are shown in Fig. 1.2

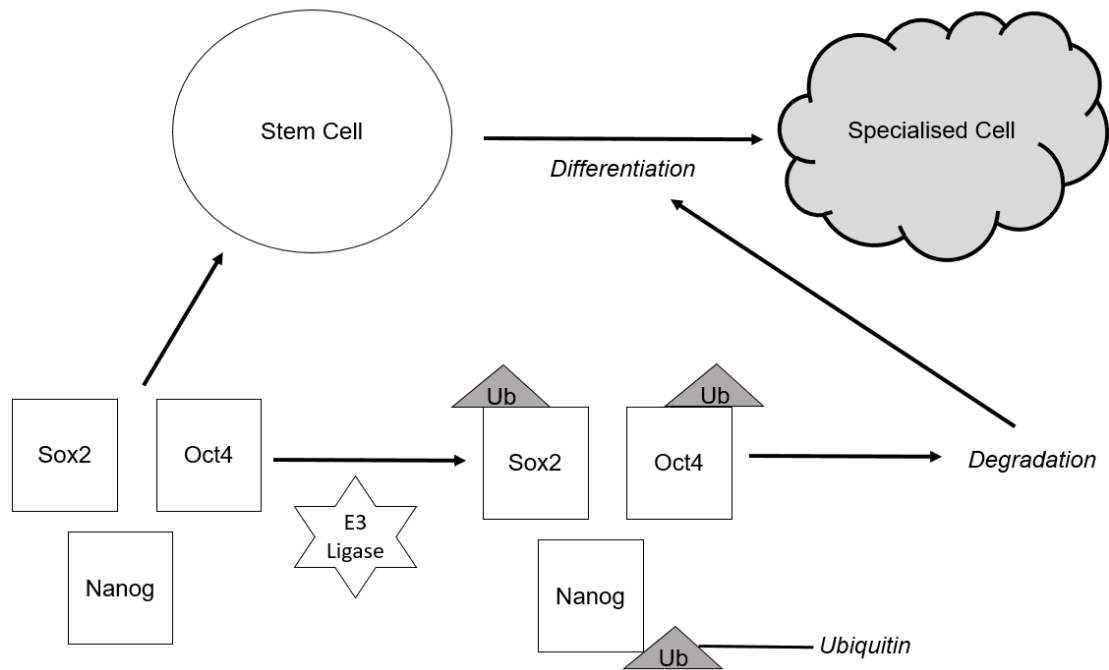


Fig. 1.2

With reference to the information and Fig. 1.2 above, describe the function of E3 ligase in the differentiation of stem cells.

- 1. E3 ligase is complementary to ubiquitin and Sox2, Oct4 and Nanog,**

allowing them to bind, ubiquitin will be bonded to Sox, Oct4 and Nanog;

- 2. Which will then be targeted for degradation by proteasome,**

which will then allow for differentiation of stem cell to occur;

[2]

Fig. 1.3 shows the effect of increasing substrate protein concentration on the rate of ubiquitination. All other variables were controlled.

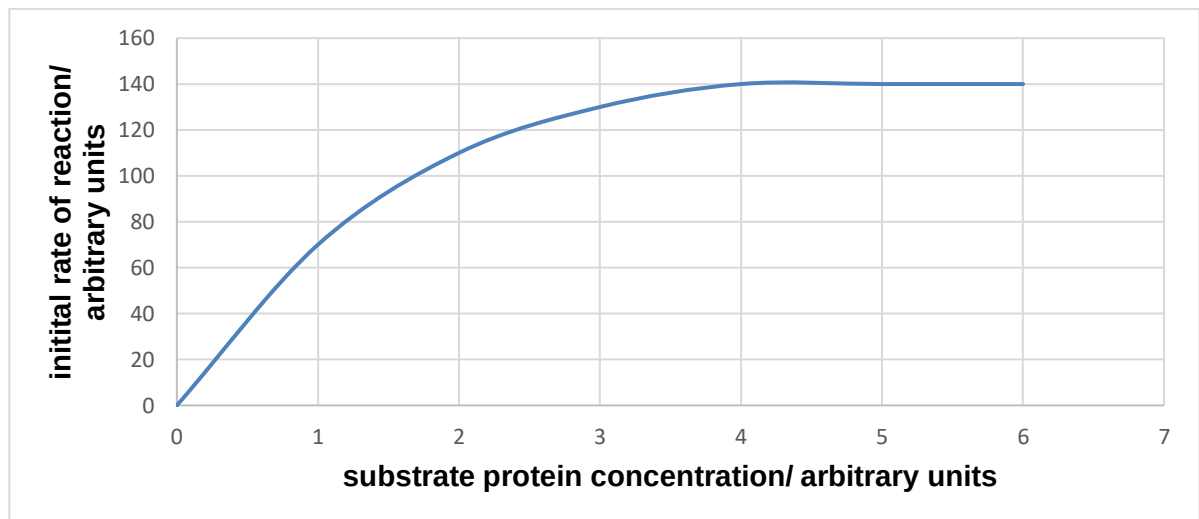


Fig. 1.3

(d) With reference to Fig. 1.3,

(i) describe the effect of increasing substrate protein concentration on activity of E3 ligase.

1. increasing substrate concentration, increases initial rate of reaction rapidly at low substrate concentration,

from 0 arbitrary units (a.u.) to about 140 a.u. when substrate concentration increased from 0 a.u. to 4 a.u;

2. then initial rate of reaction reaches a maximum & plateaus off,

at 140 a.u. when substrate concentration increases from 4 a.u. to 6 a.u;

[2]

(ii) explain the effect.

1. at low substrate concentrations, not all enzyme's active site are occupied;

with increasing substrate concentrations, increased frequency of effective collisions between enzyme & substrate molecules → increased rate of formation of ES complexes per unit time → higher rate of reaction;

2. at low substrate concentrations, substrate concentration is the limiting factor;

at high substrate concentrations, with increasing substrate molecules, all enzyme's active site are occupied by substrates at any one time/ enzyme active sites are saturated with substrates;

3. maximum number of ES complexes formed per unit time → maximum rate of reaction

substrates will have to wait for available enzymes active sites to form new ES complexes;

[3]

(e) State the limiting factor when the graph in Fig. 1.3 plateaus.

temperature/ enzyme concentration/ pH;

[1]

[Total: 14]

2 Fig. 3.1 shows a segment of a DNA molecule.

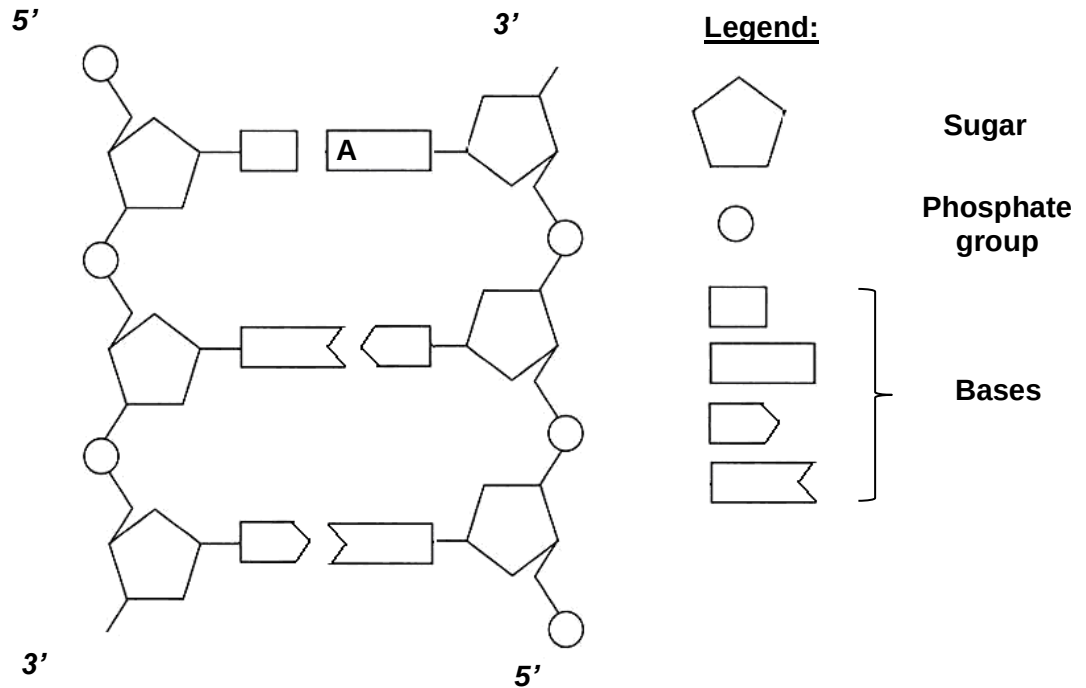


Fig. 2.1

(a) Identify the following bases:

thymine;

guanine;

[2]

(b) State the type of pentose sugar present in this molecule.

deoxyribose;;

[1]

(c) On Fig. 2.1, label the 5' and 3' ends on **both** polynucleotide strands.

refer to Fig 2.1. ® if label only on ends of one strand/ one end of both strands;

[1]

(d) Describe **two** structural differences between DNA and RNA.

1. pentose sugar in DNA is deoxyribose while in RNA is ribose;

2. DNA is usually double-stranded while RNA is usually single- stranded;

3. thymine base is present in DNA while uracil base is present in RNA;

4. shape of DNA is a double helix while for RNA, it can be linear (mRNA) or clover-shaped (tRNA);

[2]

(Any 2 pts)

- (e) The DNA molecule is synthesised via a process called DNA replication.

Explain the role of DNA polymerase III in this process.

1. recognizes and binds to free 3'OH end of RNA primer,

elongate daughter DNA strand by adding deoxyribonucleotides complementary to parental template strand;

2. catalyse formation of phosphodiester bonds between deoxyribonucleotides,

proofreads newly synthesised strand by replacing mismatched bases with correct ones;

[2]

- (f) Fig. 2.2 shows a bacterial chromosome and a human chromosome.

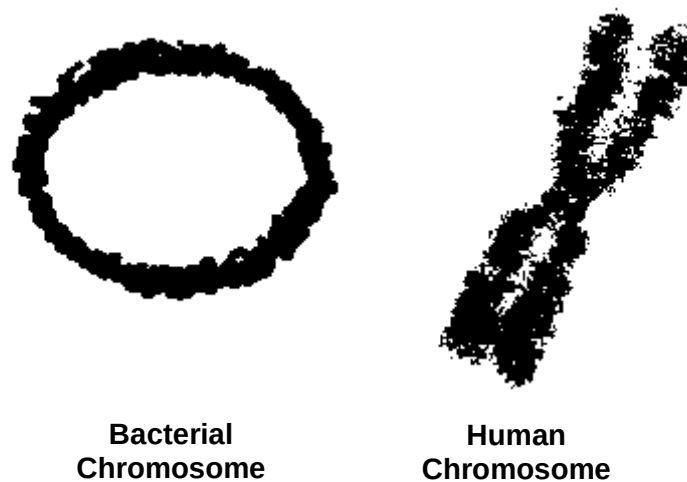


Fig. 2.2

With reference to Fig. 2.2, suggest why end-replication problem does not occur in the bacterial chromosome.

1. end-replication problem is the shortening of telomere at 5' ends of linear chromosome with every round/cycle of DNA replication,

due to absence of free 3' OH end on existing daughter strand, DNA polymerase III cannot continue to add complementary deoxyribonucleotides resulting in 3' overhang;

2. prokaryotic chromosome is circular/ do not have open ends,

presence of free 3' OH ends for DNA polymerase III to add in deoxyribonucleotides → entire chromosome can be replicated;

[2]

[Total: 10]

3 Fig. 3.1 shows the changes in DNA content in a cell during the cell cycle.

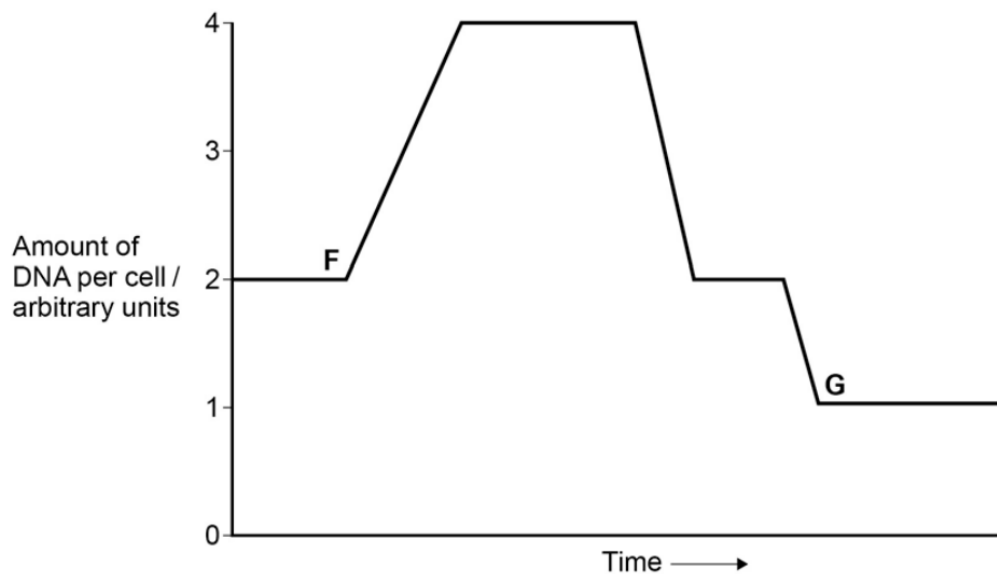


Fig. 3.1

(a) With reference to Fig. 3.1,

(i) state the type of nuclear division shown.

meiosis;

[1]

(ii) explain how the behaviour of chromosomes causes these changes in the amount of DNA per cell between F and G.

1. at F, DNA replication during S phase of interphase

results in doubling of DNA content from 2 to 4 arbitrary units (a.u.);

2. separation of homologous chromosomes during anaphase I

resulting in haploid nucleus at end of telophase I from 4 to 2 a.u.;

3. at G, separation of sister chromatids during anaphase II

resulting in decrease in DNA content from 2 to 1 a.u.;

[3]

(iii) explain the significance of the process in (a)(i).

1. meiosis produces haploid daughter cells

with half the chromosome number of the parent / from diploid parent cells;

2. to restore diploid state

upon fusion of gametes during fertilisation;

[2]

Microtubules are made from polymerisation of tubulin dimers. Fig. 3.2 shows the structure of the tubulin dimer.

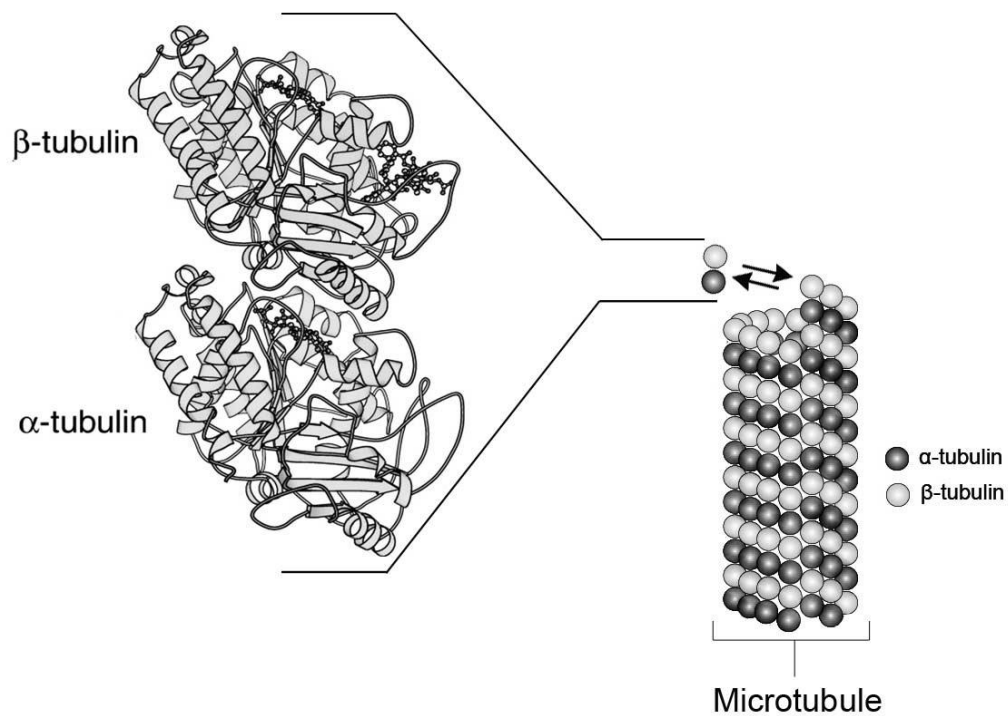


Fig. 3.2

(c) With reference to Fig. 3.2 and your knowledge of level of organisation of protein structure, describe the structure of tubulin.

1. polypeptide chain is folded to form secondary structures α -helix and β -pleated sheet

held by H bonds between -CO and -NH of peptide bonds;

2. which are further folded to form globular proteins α -tubulin and β -tubulin

held by R group interactions;

3. quaternary structure made up of α -tubulin and β -tubulin

held by R group interactions;

[3]

- (d) Tubulin inhibitors like paclitaxel and vinblastine have been utilised in chemotherapy drug trials to treat cancers. Tubulin inhibitors are known to bind to the β -tubulin subunit.

Suggest how tubulin inhibitors reduce tumour formation.

1. tubulin inhibitors interfere with / prevent polymerisation of tubulin dimers

prevent formation of spindle fibres;

2. $\therefore$ unable to bind to chromosomes

➡ chromosomes unable to separate;

3. prevent mitosis from occurring

prevent uncontrolled cell division;

[3]

[Total: 12]

- 4 *Streptococcus pneumoniae* is a bacterium normally present in the nose and throat of healthy individuals. However, it can cause serious diseases such as pneumonia (infection of lungs) and bacterial meningitis (infection of the membrane around the brain).

(a) Describe how the bacterial DNA is packaged to fit into the nucleoid region of a *S. pneumoniae* cell.

1. Circular double-stranded DNA molecule

associated with a small amount of non-histone proteins;

2. The associated (non-histone) proteins cause the DNA to coil tightly to form loop domains,

and subsequently supercoil to form condensed DNA / bacterial chromosome;

[2]

(b) Antibiotic resistance in *S. pneumoniae* is a major concern for clinicians.

Explain how an antibiotic resistant strain of *S. pneumoniae* could have transferred the antibiotic resistance gene to normal *S. pneumoniae* that is present in the throat of a healthy adult.

Either:

1. Bacterial conjugation,

direct transfer of plasmid DNA from antibiotic resistant bacterium to normal bacterium in the throat;

2. Plasmid DNA codes for sex pilus,

involved in the formation of a cytoplasmic bridge between the two bacteria to facilitate transfer of the antibiotic resistance gene (between cells);

3. Single-stranded F plasmid DNA containing antibiotic resistance gene transferred from antibiotic resistant donor bacterium

to normal recipient cell and replicated to form double-stranded DNA;

Or:

4. Bacterial transformation,

Antibiotic resistant bacteria died / lysed (in the throat) and DNA is released;

5. Competent *S. pneumoniae* / *S. pneumoniae* with competence factors

take up DNA containing antibiotic resistance gene from the environment;

6. Homologous recombination occur

so that antibiotic gene is incorporated into the bacterial DNA;

[3]

- (c) To eradicate the harmful bacteria, some scientists researched on developing a modified virus that could destroy the bacteria.

Describe and explain how the reproductive cycle of such viruses would be able to destroy the bacteria without harming humans.

1. Modified T4 bacteriophages / phages that undergo lytic cycle,

infect bacteria by binding to specific receptors on the cell surface;

2. Phage injects viral DNA into host cell;

viral DNA codes for enzymes that shut down metabolic activities of the host and degrades host's cell DNA;

3. Phage replicates its genome and synthesises its proteins using host's cell metabolic machinery / enzymes and ribosomes, phage-coded lysozyme lyse the bacteria cell wall to release assembled phages from the bacteria to infect other bacteria;

4. Human cells not having the receptors that allow attachment of phages;

[4]

- (d) A certain bacteriophage is known to insert its own genome between the *lacZ* and *lacY* structural genes of the *lac* operon in *S. pneumoniae*, to give prophages.

Explain the change in number of bacteriophage, if any, when host bacteria are grown in environment with an abundant supply of glucose and no lactose.

1. no allolactose binds to allosteric site of repressor,

thus repressor remains bound to operator region and the operon is not switched on;

2. no transcription of structural genes and viral genes by RNA polymerase as promoter sequence is blocked by repressor,

hence no synthesis of phage products, thus no new phage particles formed;

[2]

[Total: 11]

- 5 Wiskott-Aldrich syndrome is an example of rare disorders in which there is a defect in one or more aspects of the immune system. In Wiskott-Aldrich syndrome, the functioning of B lymphocytes and T lymphocytes is defective and there is a low platelet count, with the size of platelets being smaller than normal.

- (a) A study of six generations of one family led to the conclusion that the Wiskott-Aldrich syndrome showed X-linked inheritance.

Grace who is normal, has a father who is normal and a brother who died of Wiskott-Aldrich syndrome when he was three years old. She marries Joshua who does not have Wiskott-Aldrich syndrome. The couple decides to have a child.

It was later discovered that both are also heterozygous for the gene coding for the enzyme tyrosinase which catalyses the first two steps in the conversion of tyrosine to melanin, the major pigment found in melanocytes. Albinism is an autosomal recessive disorder which affects the production of melanin.

Using the following symbols,

X^A: dominant allele for no Wiskott-Aldrich syndrome on X chromosome

X^a: recessive allele for Wiskott-Aldrich syndrome on X chromosome

N: dominant allele for normal melanin production

n: recessive allele for albinism

draw a genetic diagram, explain the probability that their first son has both Wiskott-Aldrich syndrome and albinism.

Accept EITHER:

Parental phenotype:

Normal melanin production,
no Wiskott-Aldrich syndrome

male

Normal melanin production,
No Wiskott-Aldrich syndrome
female

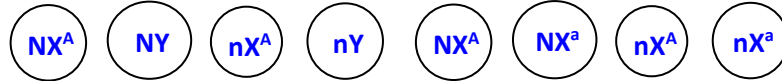
Parental genotype
[1]:

NnX^AY

×

NnX^AX^a

Gametes [1]:



Fertilisation [1]:

Gametes	NX^A	NX^a	nX^A	nX^a
NX^A	NNX^AX^A	NNX^AX^a	NnX^AX^A	NnX^AX^a
NY	NNX^AY	NNX^aY	NnX^AY	NnX^aY
nX^A	NnX^AX^A	NnX^AX^a	nnX^AX^A	nnX^AX^a
nY	NnX^AY	NnX^aY	nnX^AY	nnX^aY

F₁ genotypic ratio:

$\underbrace{3 N_X^AX^A : 3 N_X^AX^a : 3 N_X^AY : 3 N_X^aY : 1 nnX^AY : 1 nnX^aY}_{\text{F}_1 \text{ genotypic ratio}}$

F₁ phenotypic ratio:

6 : 2 : 3 : 3 : 1 : 1
 Normal melanin production, no Wiskott-Aldrich syndrome female : Albino, no Wiskott-Aldrich syndrome female : Normal melanin production, no Wiskott-Aldrich syndrome male : Normal melanin production, Wiskott-Aldrich syndrome male : Albino, no Wiskott-Aldrich syndrome male : 1 albino, Wiskott-Aldrich syndrome male

Correct F₁ genotypic and corresponding phenotypic ratio [1]

Probability that 1st child is an albino male with Wiskott-Aldrich syndrome:

1/16 or 6.25% or 0.0625 [1]

OR:

Parental phenotype: Normal melanin production, no Wiskott-Aldrich syndrome male Normal melanin production, No Wiskott-Aldrich syndrome female

Parental genotype NnX^AY $\times$ NnX^AX^A
[1]:

Gametes [1]: NX^A NY nX^A nY NX^A nX^A

Fertilisation [1]:

Gametes	NX^A	NY	nX^A	nY
NX^A	NNX^AX^A	NNX^AY	NnX^AX^A	NnX^AY
nX^A	NnX^AX^A	NnX^AY	nnX^AX^A	nnX^AY

F₁ genotypic ratio:

3 $N_X^AX^A$: 1 nnX^AX^A : 3 N_X^AY : 1 nnX^AY :

F₁ phenotypic ratio:

3 : 1 : 3 : 1
Normal melanin production, no Wiskott-Aldrich syndrome female Albino, no Wiskott-Aldrich syndrome female Normal melanin production, no Wiskott-Aldrich syndrome male Albino, No Wiskott-Aldrich syndrome male

Correct F₁ genotypic and corresponding phenotypic ratio [1]

Probability that 1st child is an albino male with Wiskott-Aldrich syndrome: 0 or 0% [1]

[5]

In humans, two genes control the presence of a rare form of diabetes mellitus. These two genes, **R/r** and **E/e**, code for the insulin receptor tyrosine kinase (RTK) and insulin receptor substrate 1 (IRS1) respectively.

In high blood glucose concentrations, insulin binds to RTK, initiating a series of responses that result in the activation of IRS1. Activated IRS1 then phosphorylates other relay proteins, transducing the signal into a variety of cellular responses.

In a case study, the phenotypes of offspring from double heterozygotic parents were traced and recorded.

- | | |
|----------|----|
| normal | 91 |
| diabetic | 68 |
- (b) (i)** Identify the type of interaction between the gene loci
- complementary gene action / duplicate recessive epistasis;** [1]
-
- (ii)** Explain how you arrived at your answer in part (i).
- 1. presence of both dominant alleles R and E ➡ normal**
-
- dominant alleles R & E code for functional form of RTK & IRS1;**
-
- 2. insulin able to bind to RTK ➡ phosphorylation / activation of IRS1,**
-
- IRS1 able to initiate phosphorylation cascade ➡ cellular response to ↓ blood glucose concentration;**
-
- 3. absence of dominant allele at either locus / homozygous recessive at either locus ➡ diabetic,**
-
- recessive alleles r & e code for non-functional RTK & IRS1**
-
- ➡ unable to respond to insulin secretion;** [3]
-

From the data observed, a geneticist hypothesised that inheritance of the rare form of diabetes mellitus follows the expected phenotypic ratio of 9 normal: 7 diabetic.

A χ^2 test was carried out and a calculated value of 0.0624 was obtained.

Part of the critical values of the chi-squared distribution is shown in Table 5.1 below.

Table 5.1

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

- (c) Using Table 5.1 and the calculated χ^2 value, explain the conclusion that can be drawn regarding the geneticist's hypothesis.

1. At 1 df, calculated χ^2 value of 0.0624 is < critical value of 3.84 at P = 0.05;

2. Reject the null hypothesis,

there is no significant difference between the numbers of expected and observed phenotypes;

3. Deviation is due to chance alone,

the experimental results follows the expected 9:7 phenotypic ratio;

[3]

[Total: 12]

6 Fig. 6.1 shows a process occurring during aerobic respiration.

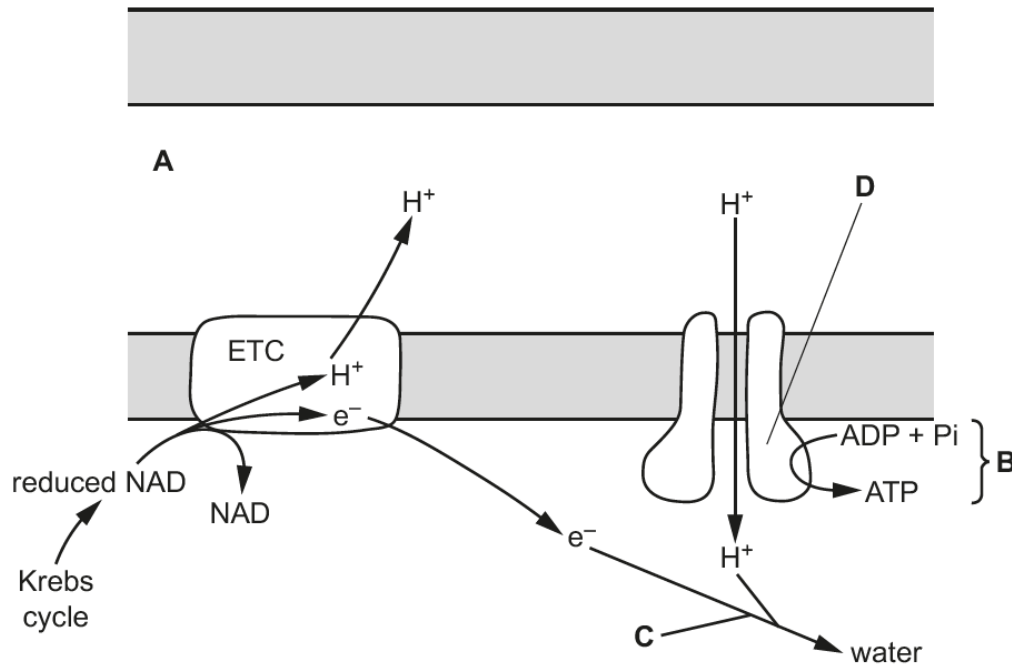


Fig. 6.1

(a) With reference to Fig. 6.1,

(i) describe the role of molecule C in accumulation of protons at region A.

1. act as final electron acceptor, accept e^- and combine with H^+ to form water
catalysed by cytochrome oxidase ;

2. maintain flow of electrons

which are passed down e^- carriers of progressively lower energy level ;

3. release energy to pump H^+ from mitochondrial matrix

to intermembrane space ;

[3]

(ii) describe the role of molecule D in process B.

1. provide hydrophilic channel for diffusion of H^+

from intermembrane space to mitochondrial matrix ;

2. via ATP synthase

phosphorylating ADP to form ATP ;

[2]

- (b) During anaerobic respiration, molecule **C** is absent.

Briefly describe how muscular contractions in an athlete can continue in the absence of molecule **C**.

1. during glycolysis, glucose is broken down to pyruvate

(net) 2 ATP is produced via substrate level phosphorylation which is used for muscular contraction ;

2. (in absence of oxygen), lactate fermentation occurs where pyruvate is converted to lactate

regenerating NAD to sustain glycolysis ;

[2]

An experiment was conducted to investigate respiration in small aerobic crustaceans. Fig. 6.2 shows the two experimental set-ups, **P** and **Q**.

The distance moved by the coloured liquid marker in the capillary tube was used to determine the volume of gases exchanged during respiration. Soda-lime, which absorbs carbon dioxide, was added to set-up **P**.

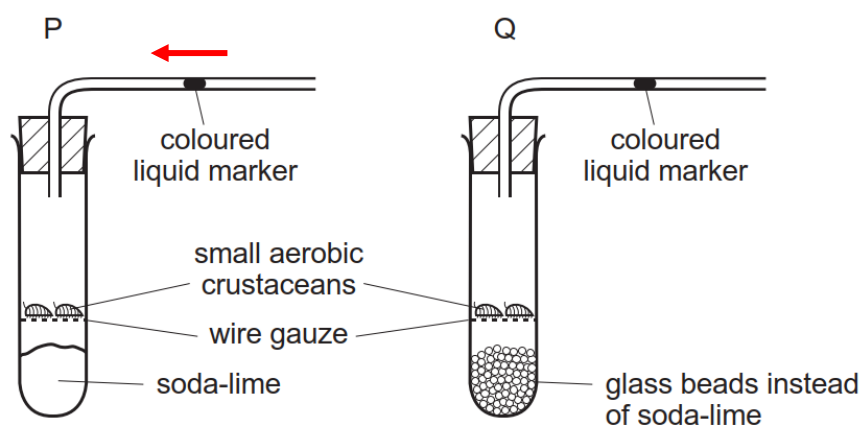


Fig. 6.2

Table 6 shows the results obtained.

Table 6

set-up	distance moved by coloured liquid marker / mm
P	8
Q	3

- (c) (i) On Fig. 6.2, use an arrow to indicate the direction of movement of the coloured liquid marker in set-up **P**. [1]

- (ii) Using your knowledge of the processes involved in aerobic respiration, explain your answer in (c)(i).

1. carbon dioxide given out during (oxidative decarboxylation reactions in) Krebs cycle

absorbed by soda lime ;

2. oxygen taken up (as final electron acceptor) during oxidative phosphorylation

causes coloured liquid marker to move to the left ;

[2]

- (iii) The respiratory quotient (RQ) is the ratio of the volume of carbon dioxide produced to the volume of oxygen consumed by the body per unit time. It can be used to give an indication of the respiratory substrate used by an organism.

$$\text{RQ} = \frac{\text{volume of carbon dioxide produced}}{\text{volume of oxygen consumed}}$$

Using the information from Table 7, as well as the following information, calculate the respiratory quotient.

diameter of capillary tube = 0.4 mm

area of circle = πr^2

$\pi = 3.14$

distance moved due to O₂ = 8 mm

distance moved due to O₂ and CO₂ = 3 mm

distance moved due to CO₂ = 8 - 3 = 5 mm

volume of CO₂ = 3.14 x 0.2² x 5 mm³

volume of O₂ = 3.14 x 0.2² x 8 mm³;

RQ = volume of O₂ / volume of CO₂ = 0.625 ;

[2]

[Total: 12]

- 7 A subspecies is a genetically distinct population within a species that has some phenotypic differences from the rest of the species, but is not yet reproductively isolated.

Nine subspecies of the tiger, *Panthera tigris*, have been identified. Six of these subspecies are found on mainland Asia. Three of the subspecies originate from the Sunda Islands. These islands include Bali, Java and the large island of Sumatra.

Fig. 7.1 shows these three islands.

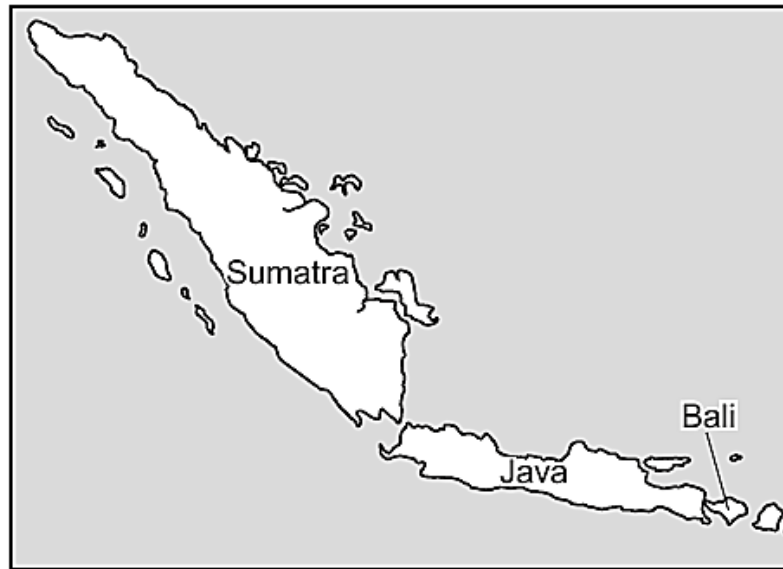


Fig. 7.1

- The Bali tiger, *Panthera tigris balica* (*P. t. balica*), became extinct in the 20th Century. The Bali tiger was found only on the island of Bali.
- The Javan tiger, *P. t. sondaica*, became extinct in the 20th Century. The Javan tiger was found only on the island of Java.
- The Sumatran tiger, *P. t. sumatrae*, lives only on Sumatra and is the closest living relative of Bali and Javan tigers.

20 000 years ago land bridges temporarily connected the Sunda Islands.

(a) Suggest **and** explain how the three subspecies of tiger on the Sunda Islands formed.

1. different environmental conditions on different islands act as selection pressures;

select for tigers with different favourable phenotypes;

2. these tigers experience higher survival rate & reproductive success,

thus different favourable alleles are passed down to offspring resulting in different change in allelic frequency;

3. populations also accumulate different mutations,

causing genetic divergence between the three populations leading to descent with modification;

4. populations cannot interbreed due to geographical barrier resulting in no gene flow,

leading to genetic divergence of gene pools;

5. However genetic differences distinctive enough to classify them as subspecies, are insufficient to result in formation of reproductive barrier due to insufficient time to accumulate mutations in each isolated population,

as land bridge that joins the Sunda islands is broken only 20000 years ago thus the 3 populations are not considered as separate species;

[5]

A recent study carried out a genetic analysis of the nine subspecies of tiger. Specific sections of mitochondrial DNA (mtDNA) were amplified using PCR and compared to assess their evolutionary history.

- The source of DNA for the extinct subspecies came from museum specimens.
- The mtDNA sections for the three island subspecies were genetically distinct from the other six mainland subspecies.

(b) Describe **one** characteristic of mtDNA that makes it more useful than using nuclear DNA to provide evidence of evolution.

Any 1 below:

1. large quantity of mtDNA in the cell,

so easier to extract / amplify DNA for testing;

2. small genome size of mtDNA,

so easier to locate specific region of DNA for testing;

3. mtDNA present in a single copy,

so change in mtDNA sequence is solely result of mutation;

4. mtDNA is inherited maternally / from the mother,

so all mtDNA sections are shared between all members (of maternal) family;

[1]

- (c) Different pairs of primers are used to amplify different regions of the mtDNA from the nine subspecies.

Explain why a specific pair of primer is used for amplification of a specific region of the mtDNA. .

(bind to region flanking target sequence on mtDNA) to provide free 3'OH

for recognition by DNA polymerase for synthesis of daughter strand;

[1]

- (d) Suggest why *P. t. sumatrae* is still considered to be a member of the species *Panthera tigris*.

1. *P. t. sumatrae* is morphologically / physiologically similar / similar in behaviour to mainland tigers / *Panthera tigris*,

also genetically similar to mainland tigers / *Panthera tigris*;

2. still able to interbreed with mainland tigers / *Panthera tigris*

to produce viable and fertile offspring;

[2]

- (e) State the genus of the Bali tiger.

Panthera

[1]

Fig. 7.2 shows the phylogenetic tree of big cats constructed based on nucleotide sequences obtained from fossil and biogeographical data.

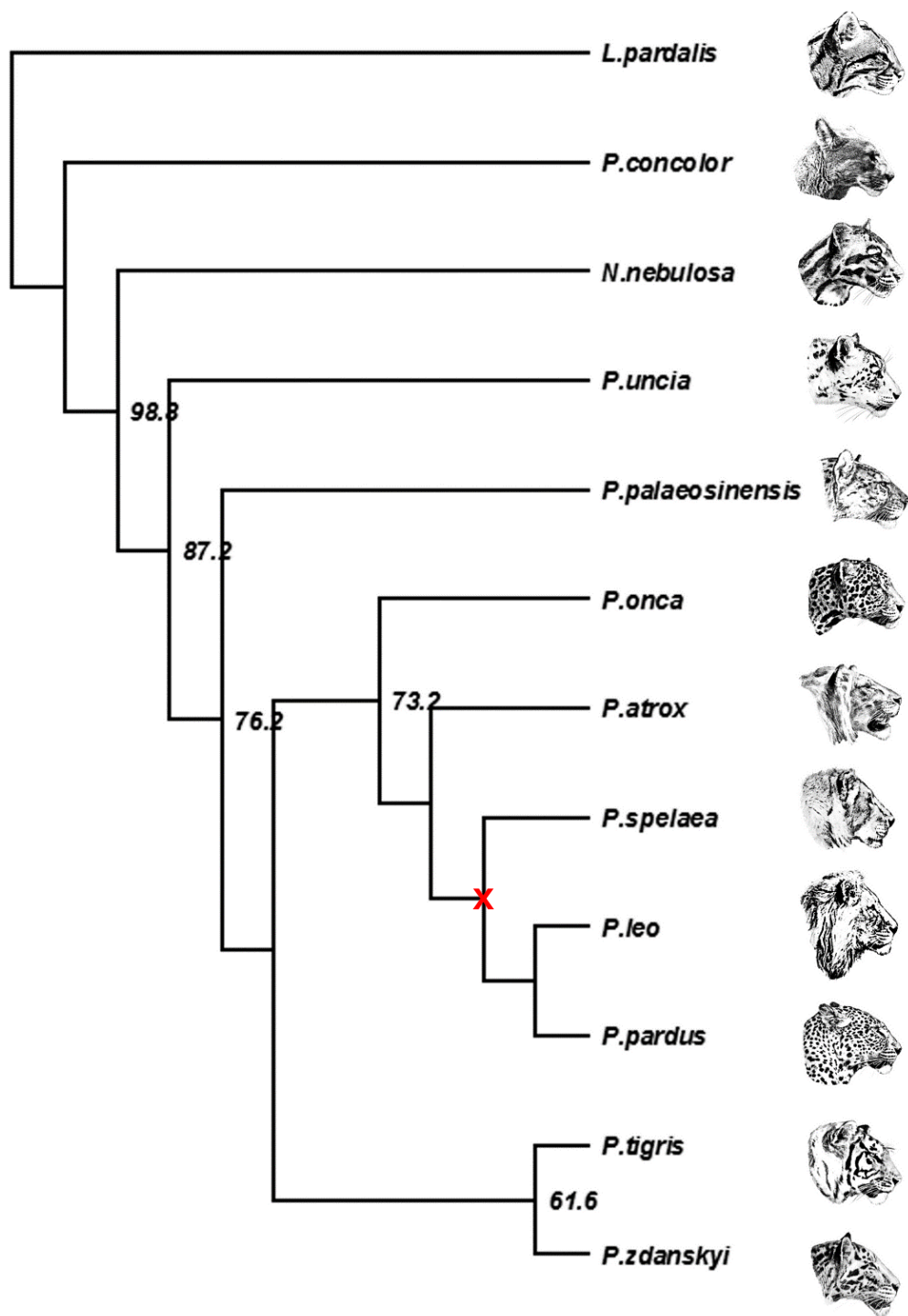


Fig. 7.2

- (f) (i) Explain how nucleotide sequences can be used in reconstructing phylogenetic relationships.
1. Compare DNA nucleotide sequence of a common gene among the organisms,

 fewer differences/ more similarities indicate that individuals share a more recent common ancestor;

 2. and have diverged more recently

 thus have accumulated mutations over a shorter period of time ; [2]

- (ii) Describe **one** advantage of using method in (e)(i).
1. objective & unambiguous

 as all organisms have 4 universal bases, A, T, C, G;

 2. precise, accurate & quantifiable

 data can be subjected to statistical analysis thus able to infer degree of relatedness between organisms;

 [1]

- (iii) Label with an 'X' in Fig. 7.2, the most recent common ancestor of *P. spelaea* and *P. leo*. [1]

[Total: 14]

8 Cry toxins are present in insecticides.

One way in which cry toxin causes cell death in target insects is by oligomerising into pore complexes in cell surface membrane of mid-gut epithelial cells.

- (a) Explain how formation of pores in the cell surface membrane of mid-gut epithelial cells causes death in target insects.

1. The pore complexes allow ions and water to diffuse freely into the cells

causing changes in membrane permeability,

2. leading to cell swelling/ cell lysis and death of mid-gut epithelial cells in target

Insects;

[2]

- (b) Another mechanism of action of Cry toxin in mid-gut epithelial cells of target insects has been proposed. Fig. 8.1 shows the proposed magnesium ion (Mg^{2+})-dependent signalling pathway.

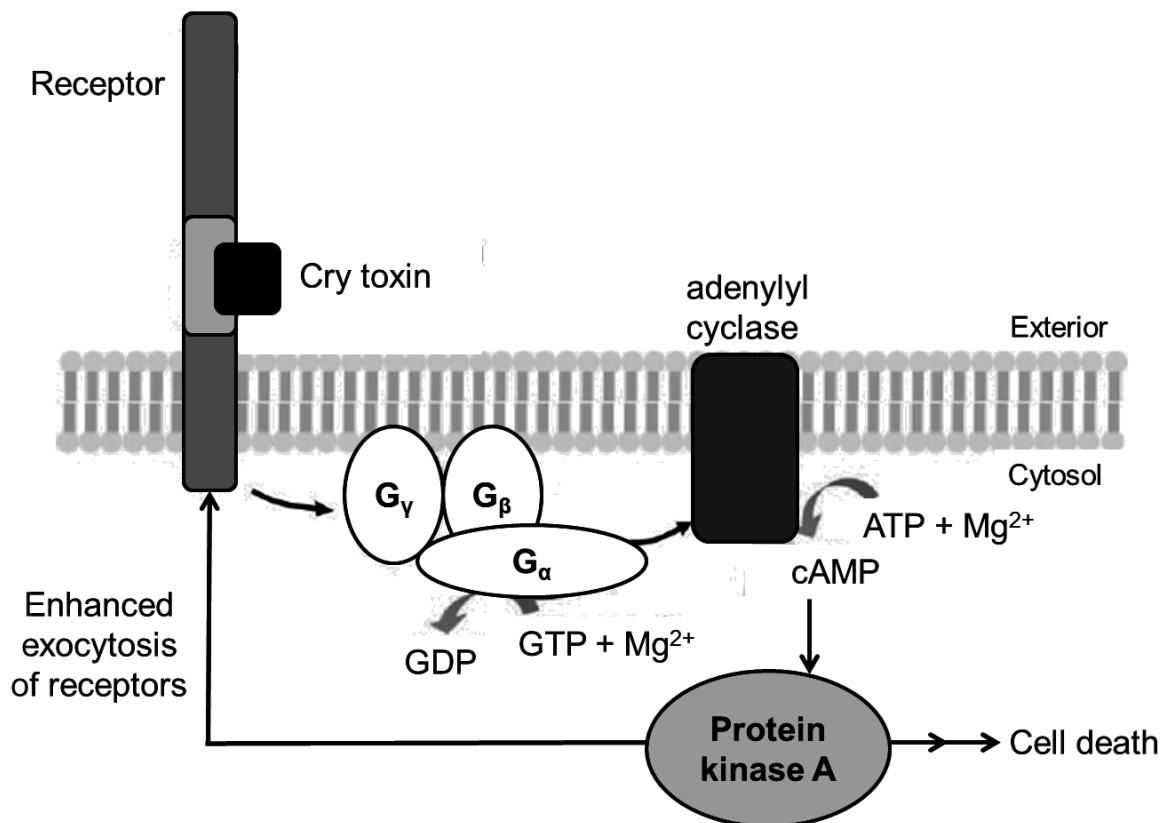


Fig. 8.1

- (i) With reference to Fig. 8.1, describe the proposed cell signalling pathway of Cry toxin.

1. When Cry toxin (ligand) binds to the extracellular domain of receptor,

the receptor is activated and cause the membrane-bound G-protein (at the cytoplasmic end) to exchange its GDP with a GTP in the presence of Mg^{2+} ;

2. α -subunit of G protein / G_α , dissociates from its β - and γ - subunits

and moves along the membrane to adenylyl cyclase;

3. Upon binding of G_α subunit, adenylyl cyclase undergoes conformational change

and catalyses conversion of ATP to cyclic adenosine monophosphate (cAMP) in the presence of Mg^{2+} ;

4. cAMP acts as a second messenger to activate relay protein, protein kinase A (PKA),

phosphorylating many downstream effector proteins and enzymes in a phosphorylation cascade

5. to induce incorporation of more receptors of Cry toxin at cell surface membrane and to promote cell death.

[5]

- (ii) NF449 is a drug that prevents binding of GTP to G protein.

Suggest how NF449 could affect the signalling pathway of Cry toxin.

1. GDP of G protein is not exchanged for GTP inactivating G protein and

adenylyl cyclase is not activated leading to a low level of cAMP;

2. PKA is inactivated causing reduced cytotoxicity of Cry toxins/ fewer cell death.

[2]

- (c) Suggest why Cry toxins only kill target insects and not other organisms.

1. Membrane receptors that bind to Cry toxins are only expressed in the mid-gut epithelium cells of target insects and not in that of other organisms;

OR

2. Conditions of gastrointestinal tract such as pH and ion balance that facilitate activation of Cry signalling cascade in insects are different in other organisms.

[1]

[Total: 10]

- 9 Whooping cough is a disease that affects some infants. Doctors collected data relating to whooping cough from 1965 to 1995.

They collected data for:

- the number of cases of whooping cough reported
- the percentage of infants vaccinated against whooping cough.

Fig. 9.1 shows the data collected by the doctors.

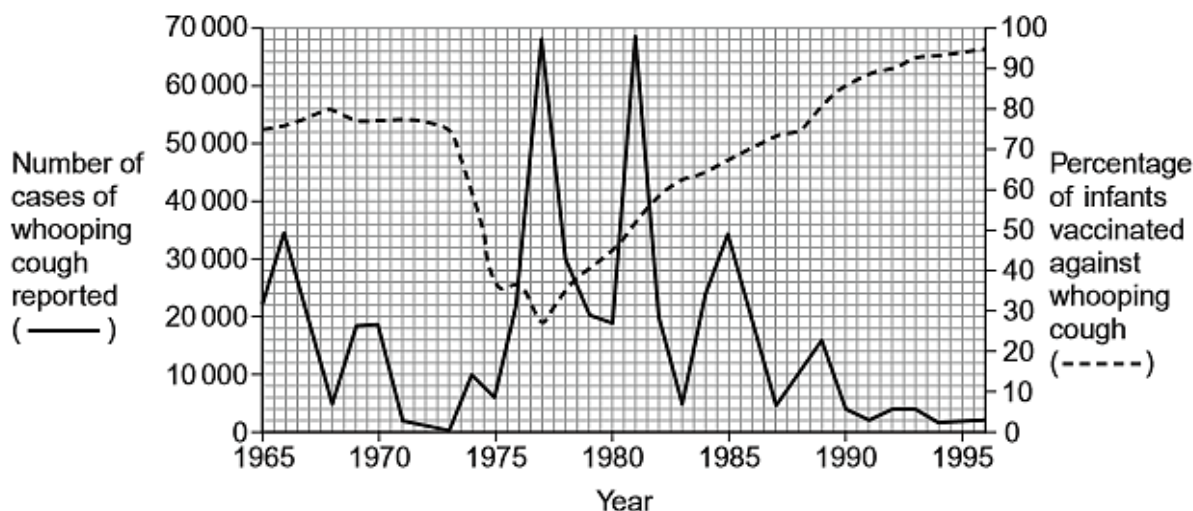


Fig. 9.1

(a) Using information from Fig. 9.1, suggest

- (i) why the percentage of infants vaccinated decreased between 1973 and 1975.

1. ↓ % vaccinated from 74% to 37% ;

2. in 1973, no. of whooping cough cases were at 0,

leading to population complacency ∴ parents did not follow immunisation schedule to bring their infants for vaccination / OWTTE ;

[2]

- (ii) why each successive peak in the number of reported cases of whooping cough decreased between 1980 and 1990.

1. ↓ no. of whooping cough cases from 69000 in 1981, to 34000 in 1985, to 15000 in 1989 ;

2. ↑ % number of vaccinations from 45% in 1980 to 85% in 1990 ;

3. higher number of individuals vaccinated against the bacteria

herd immunity achieved where bacteria does not have sufficient hosts to complete its replication and spread / transmission cycle ;

[3]

[Total: 5]